

Programmable Interval Timer for Live Shows

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Presented here is a microcontroller-based special-purpose programmable interval timer for live shows and competitions to record time while the contestants are performing. The main objective of this project is to design an accurate programmable interval timer that can be utilised in inter-college/inter-zonal/inter-university shows and competitions.

In live competitions and shows, each item has a predetermined fixed minimum time, maximum time and grace time. The item being presented should complete between minimum and maximum time. Sometimes, grace time for a few seconds over and above the maximum time is allowed. If it is not completed within the stipulated timeframe, the item presented by that particular team is disqualified.

One may use a stopwatch to monitor the competition but there are chances of inaccuracy or introduction of errors while switching on the stopwatch in the beginning or switching off at the end of a particular item. The audience or the participants can object or raise a complaint for inaccuracy in time allotment. This project comes handy to overcome these problems. It is fully automatic and programmable. It has the flexibility to program the minimum, maximum and grace time period in the database as explained in the software section. The author's prototype is shown in Fig. 1.

This project has two parts, namely, hardware and software.

Circuit and working

The circuit diagram of programmable interval timer is shown in Fig. 2. It comprises PIC16F877A microcontroller (IC3), MAX232 converter/driver (IC4), three BC547 transistors (T1, T2 and T3),

16x2 LCD (LCD1), 230V AC primary to 15V, 500mA secondary transformer (X1), bridge rectifier module (BR1), regulator ICs 7812 (IC1) and 7805 (IC2) and some other components.

PIC16F877A. This microcontroller is the heart of the hardware section. It has an 8-bit processor with 8k (14-bit words) Flash program memory, 368 bytes data memory and 256 bytes EEPROM. In addition to this, it has three timers, namely, Timer0, Timer1

and Timer2. By using an 8-bit Timer2 having a period register (PR2) and post-scalar ratio as 1:10, interrupt frequency of 100Hz was generated.

For every interrupt, the value of timer variable is increased by one. One-second time elapses when the value of variable reaches 100. The value of one second is the base of the project. When this one second reaches 60, one minute is generated.

Port B of the microcontroller is interfaced with 16x2 LCD module to display the values of minutes and seconds. Data pins D4 through D7 of the LCD are connected to RB0 through RB3 pins of the microcontroller, respectively. Port pins RB4 and RB5 of the microcontroller are connected to the register select (RS) and enable (EN) pins of the LCD, respectively. R/W is grounded to keep the write mode always enabled.

Port pins RC6 and RC7 of the microcontroller are configured as a universal synchronous asynchronous receiver transmitter (USART) by connecting to pins 11 and 12 of MAX232 IC, respectively. Pins 14 and 13 of MAX232 are connected to pins 2 and 3 of COM port DB9 connector, respectively for the data communication with personal computer (PC). You can also use a USB-to-serial converter to connect with the PC in case you do not have a 9-pin COM port on your PC.

Green, yellow and red bulbs (15W each) are connected to RD1, RD2 and RD3 pins of the microcontroller, respectively through three relays. When a team starts presenting an item, green bulb glows. When minimum time is over, green bulb goes off and yellow bulb goes on to indicate that maximum time count started.

As soon as the maximum time is



Fig. 1: Author's prototype

PARTS LIST

Semiconductors:	
IC1	- 7812, 12V regulator
IC2	- 7805, 5V regulator
IC3	- PIC16F877A microcontroller
IC4	- MAX232 driver/converter
T1-T3	- BC547 npn transistors
D1-D3	- 1N4007 rectifier diode
BR1	- 1A bridge rectifier module
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R6	- 10-kilo-ohm
R2	- 1.2-kilo-ohm
R3-R5	- 1-kilo-ohm
R7	- 100-ohm
VR1	- 10-kilo-ohm preset
Capacitors:	
C1	- 470 μ F, 35V electrolytic
C2, C3, C6-C9	- 0.1 μ F ceramic disk
C4, C5	- 27pF ceramic disk
C10	- 1 μ F, 16V electrolytic
Miscellaneous:	
X1	- 230V AC primary to 15V, 500mA secondary transformer
X _{cr} , 1	- 16MHz crystal oscillator
S1, S2	- Tactile switch
RL1-RL3	- 12V, 1/CO relay
CON1	- 9-pin DB9 COM port connector
CON2	- 2-pin terminal block
LCD1	- 16x2 LCD module
BL1-BL3	- 15W bulb